Lab-12 Ex2 - Pointer and Selection Sort

Implement selection sort for a character array:

1. Instead of sorting integers, the program now sorts an array of characters lexicographically (i.e. based on their ASCII values).
2. Implement a function called swap()which is used to swap the starting element and the smallest element found in the sub-array via two pointer parameters. The skeleton of the new function has been provided: *

```
void swap(char *x, char *y)
```

Suppose that there are two character variables `char a = 'A';` and `char b = 'B';` defined on the caller side. Calling `swap(&a, &b)` will swap the values stored in variables `a` and `b`, so a stores 'B' and b stores 'A' after the function call. You should call this `swap()` function for swapping a pair of elements, say `a[i]` and `a[minPos]` if `a[i]>a[minPos]`, when necessary, in the body of the function:

```
void selection_sort(char a[], int n)
```

Note:

Assume that n (the size of the input array) is at most 10, and all input values are valid (only upper-case or lower-case alphabets).

Complete the part for reading user's input:

- Read an integer n.
- Then, read n characters, separated by a space, to be sorted.

The template code handles the output part:

- Output the n characters, separated by a space, in ascending order based on ASCII code.
- No extra spaces after the last character.

Sample Output

5

d e a c b

```
a b c d e
```